
INSTRUCTIONS:

1. Write the table on the ruled side.
2. Write the reactions for all the confirmatory tests in each experiment on the unruled side.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 1

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO₄²⁻):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$\text{SO}_4^{2-} + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{Cl}^-$ white ppt.		

CONFIRMATORY TEST FOR SULPHATE (SO₄²⁻)

Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$\text{SO}_4^{2-} + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbSO}_4 \downarrow + 2\text{CH}_3\text{COO}^-$ white ppt.		

C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	No precipitate.	V group basic radical absent.
<u>DETECTION OF GROUP VI BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution + ammonium phosphate solution.	White crystalline precipitate	VI group basic radical present. Mg ²⁺ is present.
<u>CONFIRMATORY TEST FOR MAGNESIUM (Mg⁺²)</u>		
Original solution + NaOH solution.	White precipitate insoluble in NaOH solution	Mg ²⁺ is confirmed.
$\text{Mg}^{2+} + (\text{NH}_4)_2\text{HPO}_4 \rightarrow \text{Mg}(\text{NH}_4)\text{PO}_4 \downarrow$ $\text{Mg}^{2+} + 2\text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 \downarrow + 2\text{Na}^+$		

RESULT:

The given salt contains __sulphate(SO ₄ ⁻²)__ as acid radical and ____magnesium_(Mg ⁺²)____ as basic radical.
Hence the given salt is MgSO₄ – Magnesium sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 2

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS: DETECTION OF GROUP I ACID RADICALS

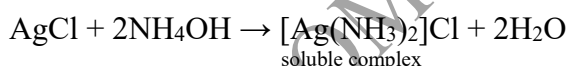
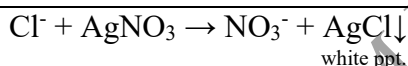
EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

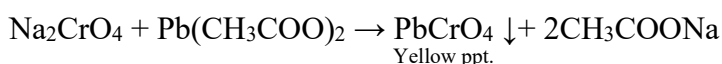
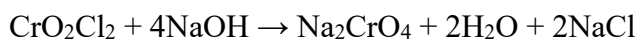
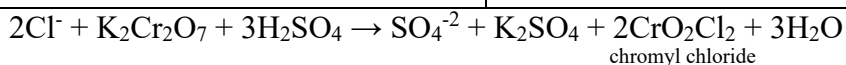
Salt + conc. H ₂ SO ₄ in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH ₄ OH.	Group II acid radical is present. Cl ⁻ may be present.
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CONFIRMATORY TEST FOR CHLORIDE (Cl⁻)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.
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2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H ₂ SO ₄ . Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.
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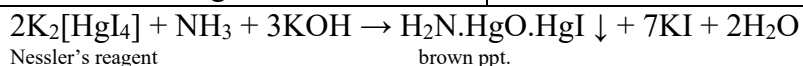
C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	A pungent smell of ammonia which gives dense white fumes with a glass rod dipped in conc.HCl.	NH_4^+ may be present.
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CONFIRMATORY TEST FOR AMMONIUM (NH₄⁺) RADICAL:

Salt solution in water + few drops of Nessler's reagent.	Brown precipitate is formed.	NH_4^+ is confirmed.
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RESULT:

The given salt contains chloride (Cl^-) as acid radical and ammonium (NH_4^+) as basic radical.

Hence the given salt is NH_4Cl – Ammonium chloride.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 3

A] PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO₄²⁻):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$\text{SO}_4^{2-} + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{Cl}^-$ white ppt.		

CONFIRMATORY TEST FOR SULPHATE (SO₄²⁻)

Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$\text{SO}_4^{2-} + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbSO}_4 \downarrow + 2\text{CH}_3\text{COO}^-$ white ppt.		

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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DETECTION OF GROUP II BASIC RADICALS

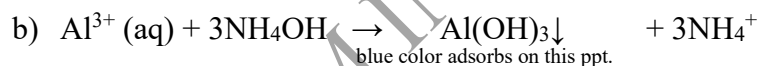
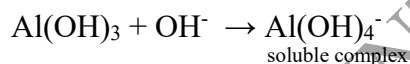
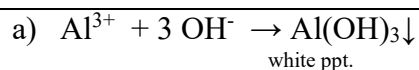
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
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DETECTION OF GROUP III BASIC RADICALS

Original Solution + NH_4Cl solid + NH_4OH solution in excess.	A gelatinous white precipitate.	III group basic radical present. Al^{3+} may be present.
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CONFIRMATORY TEST FOR ALUMINIUM(Al^{3+})

a) Original solution + NaOH solution dropwise. Above solution is treated with NH_4Cl solid. b) Lake test: Original solution + blue litmus solution + NH_4OH solution in excess.	Gelatinous white precipitate soluble in excess of NaOH. White gelatinous precipitate reappears. Bluish white precipitate, floating like lake is formed.	Al^{3+} is confirmed.
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RESULT:

The given salt contains sulphate (SO_4^{-2}) as acid radical and aluminium (Al^{+3}) as basic radical.

Hence the given salt is $\text{Al}_2(\text{SO}_4)_3$ – Aluminium sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 4

A| PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	Reddish brown fumes and the solution turns brown.	Group II acid radical is present. Br ⁻ may be present.
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CONFIRMATORY TEST FOR CHLORIDE (Br⁻)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution. $\text{Br}^- + \text{AgNO}_3 \rightarrow \text{NO}_3^- + \text{AgBr} \downarrow$ pale yellow ppt.	A pale yellow precipitate partially soluble in excess of ammonium hydroxide.	Br ⁻ is confirmed.
2.Orange globule test: Clear salt solution in water + few drops of carbon tetrachloride + Chlorine water, shaken well. $\text{Br}^- + \text{Cl}_2 \rightarrow \text{Cl}^- + \text{Br}_2$ Bromine being soluble in CCl ₄ imparts orange colour to CCl ₄ layer.	Orange brown globule separates out.	Bromide is confirmed.

C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	A pungent smell of ammonia which gives dense white fumes with a glass rod dipped in conc.HCl.	NH ₄ ⁺ may be present.
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CONFIRMATORY TEST FOR AMMONIUM (NH₄⁺) RADICAL:

Salt solution in water + few drops of Nessler's reagent.	Brown precipitate is formed.	NH ₄ ⁺ is confirmed.
$2\text{K}_2[\text{HgI}_4] + \text{NH}_3 + 3\text{KOH} \rightarrow \text{H}_2\text{N.HgO.HgI} \downarrow + 7\text{KI} + 2\text{H}_2\text{O}$ Nessler's reagent brown ppt.		

RESULT:

The given salt contains chloride (Br⁻) as acid radical and ammonium (NH₄⁺) as basic radical.

Hence the given salt is NH₄Br – Ammonium bromide.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 5

A| PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
-	+	

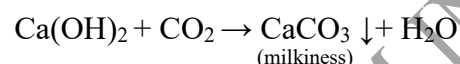
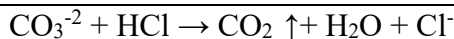
B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO ₃ ⁻²) may be present.

CONFIRMATORY TEST FOR CO₃⁻²:

Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Carbonate (CO ₃ ⁻²) is confirmed.
The above gas is passed through lime water.	The lime water turns milky.	



C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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DETECTION OF GROUP II BASIC RADICALS

Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
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<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	No precipitate.	V group basic radical absent.
<u>DETECTION OF GROUP VI BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution + ammonium phosphate solution.	White crystalline precipitate	VI group basic radical present. Mg ²⁺ is present.
<u>CONFIRMATORY TEST FOR MAGNESIUM (Mg⁺²)</u>		
Original solution + NaOH solution.	White precipitate insoluble in NaOH solution	Mg ²⁺ is confirmed.
$\text{Mg}^{2+} + \text{NH}_4\text{OH} + (\text{NH}_4)_2\text{HPO}_4 \rightarrow \text{Mg}(\text{NH}_4)\text{PO}_4\downarrow + \text{H}_2\text{O}$ $\text{Mg}^{2+} + 2\text{NaOH} \rightarrow \text{Mg}(\text{OH})_2\downarrow + 2\text{Na}^+$		

RESULT:

The given salt contains <u>carbonate(CO₃⁻²)</u> as acid radical and <u>magnesium (Mg⁺²)</u> as basic radical.
Hence the given salt is MgCO₃ – Magnesium carbonate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 6

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings.	No reaction	
Heated strongly.		

DETECTION OF GROUP III ACID RADICAL(SO₄²⁻):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$\text{SO}_4^{2-} + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{Cl}^-$ white ppt.		

CONFIRMATORY TEST FOR SULPHATE (SO₄²⁻)

Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$\text{SO}_4^{2-} + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbSO}_4 \downarrow + 2\text{CH}_3\text{COO}^-$ white ppt.		

C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original solution + NH ₄ Cl solid +NH ₄ OH solution in excess. Warm + H ₂ S	White precipitate.	IV group basic radical present. Zn ⁺² may be present.
CONFIRMATORY TEST FOR ZINC (Zn²⁺)		
Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts. i) To one part of the solution add dilute NaOH solution drop by drop till it is excess. ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.	White precipitate is soluble in excess of NaOH. White precipitate reappears. White (or bluish/grayish white) precipitate.	Zn ⁺² confirmed.
a) $\text{Zn}^{2+} + 2\text{NaOH} \rightarrow \text{Zn(OH)}_2 \downarrow + 2\text{Na}^+$ $\text{Zn(OH)}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + 2\text{H}_2\text{O}$ b) $2\text{Zn}^{2+} + \text{K}_4[\text{Fe(CN)}_6] \rightarrow \text{Zn}_2[\text{Fe(CN)}_6] \downarrow + 4\text{K}^+$		

RESULT:

The given salt contains <u>sulphate(SO₄⁻²)</u> as acid radical and <u>zinc (Zn⁺²)</u> as basic radical.
Hence the given salt is ZnSO₄ – zinc sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 7

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH ₄ OH.	Group II acid radical is present. Cl ⁻ may be present.
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CONFIRMATORY TEST FOR CHLORIDE (Cl⁻)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.
$\text{Cl}^- + \text{AgNO}_3 \rightarrow \text{NO}_3^- + \text{AgCl} \downarrow$ white ppt. $\text{AgCl} + 2\text{NH}_4\text{OH} \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{H}_2\text{O}$ soluble complex		
2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H ₂ SO ₄ . Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.
$2\text{Cl}^- + \text{K}_2\text{Cr}_2\text{O}_7 + 3\text{H}_2\text{SO}_4 \rightarrow \text{SO}_4^{2-} + \text{K}_2\text{SO}_4 + 2\text{CrO}_2\text{Cl}_2 + 3\text{H}_2\text{O}$ chromyl chloride $\text{CrO}_2\text{Cl}_2 + 4\text{NaOH} \rightarrow \text{Na}_2\text{CrO}_4 + 2\text{H}_2\text{O} + 2\text{NaCl}$ $\text{Na}_2\text{CrO}_4 + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbCrO}_4 \downarrow + 2\text{CH}_3\text{COONa}$ Yellow ppt.		

<u>Cl ANALYSIS OF BASIC RADICALS (CATIONS):</u>		
<u>TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)</u>		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
<u>DETECTION OF GROUP I BASIC RADICALS</u>		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	A white precipitate.	V group basic radical present. Ba ²⁺ , Sr ²⁺ , or Ca ²⁺ may be present.
<u>CONFIRMATORY TEST FOR BARIUM (Ba²⁺)</u>		
Original solution + acetic acid + potassium chromate solution.	Yellow precipitate.	Ba ²⁺ is confirmed.
$\text{Ba}^{2+} + 2\text{CH}_3\text{COO}^- \rightarrow (\text{CH}_3\text{COO})_2\text{Ba}$ $(\text{CH}_3\text{COO})_2\text{Ba} + \text{K}_2\text{CrO}_4 \rightarrow \text{BaCrO}_4 \downarrow + 2\text{CH}_3\text{COOK}$		
Flame test: A pinch of salt + few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Apple green colour is imparted to the flame.	Ba ²⁺ is confirmed.
<u>RESULT:</u>		
The given salt contains <u>chloride(Cl⁻)</u> as acid radical and <u>barium (Ba²⁺)</u> as basic radical.		
Hence the given salt is BaCl₂ – Barium chloride.		

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 8

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

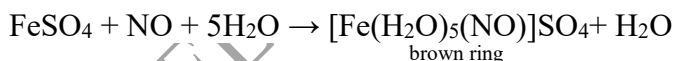
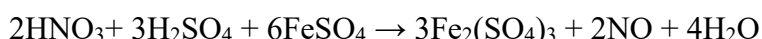
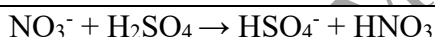
EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	II group acid radical is present. Nitrate (NO ₃ ⁻) may be present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	

CONFIRMATORY TEST FOR NITRATE(NO₃⁻):

Brown ring test: Salt solution in dil.H ₂ SO ₄ +freshly prepared saturated solution of Mohr's salt + conc.H ₂ SO ₄ added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO ₃ ⁻) is confirmed.
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C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	A white precipitate.	V group basic radical present. Ba ²⁺ , Sr ²⁺ , or Ca ²⁺ may be present.
<u>CONFIRMATORY TEST FOR BARIUM (Ba²⁺)</u>		
Original solution + acetic acid + potassium chromate solution.	No precipitate.	Ba ²⁺ is absent.
<u>CONFIRMATORY TEST FOR STRONTIUM (Sr²⁺)</u>		
Original solution + ammonium sulphate solution, warm.	White precipitate.	Sr ²⁺ is confirmed.
$\text{Sr}^{2+} + (\text{NH}_4)_2\text{SO}_4 \rightarrow \text{SrSO}_4 \downarrow + 2\text{NH}_4^+$		
Flame test: A pinch of salt + few drops conc. HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Crimson red colour is imparted to the flame.	Sr ²⁺ is confirmed.

RESULT:

The given salt contains <u>nitrate(NO₃⁻)</u> as acid radical and <u>strontium(Sr⁺²)</u> as basic radical.
Hence the given salt is Sr(NO₃)₂– Strontium nitrate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 9

A| PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
-	+	

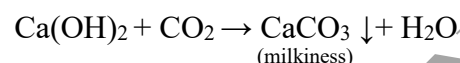
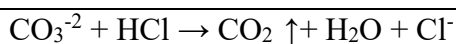
B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. H ₂ SO ₄	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO ₃ ⁻²) may be present.

CONFIRMATORY TEST FOR CO₃⁻²:

Salt + Dil. H ₂ SO ₄	Brisk effervescence is observed. A colorless gas is liberated.	Carbonate (CO ₃ ⁻²) is confirmed.
The above gas is passed through lime water.	The lime water turns milky.	



C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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DETECTION OF GROUP II BASIC RADICALS

Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
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<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	A white precipitate.	V group basic radical present. Ba ²⁺ , Sr ²⁺ , or Ca ²⁺ may be present.
<u>CONFIRMATORY TEST FOR BARIUM (Ba²⁺)</u>		
Original solution + acetic acid + potassium chromate solution.	No precipitate.	Ba ²⁺ is absent.
<u>CONFIRMATORY TEST FOR STRONTIUM (Sr²⁺)</u>		
Original solution + ammonium sulphate solution, warm.	No precipitate.	Sr ²⁺ is absent.
<u>CONFIRMATORY TEST FOR CALCIUM (Ca²⁺)</u>		
Original solution + NH ₄ OH solution + Ammonium oxalate solution.	White precipitate	Ca ²⁺ is confirmed
$\text{Ca}^{2+} + (\text{NH}_4)_2\text{C}_2\text{O}_4 \rightarrow \text{CaC}_2\text{O}_4 \downarrow + 2\text{NH}_4^+$		
Flame test: A pinch of salt + few drops conc. HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Brick red colour is imparted to the flame.	Ca ²⁺ is confirmed.

RESULT:

The given salt contains carbonate(CO₃⁻²) as acid radical and calcium (Ca⁺²) as basic radical.

Hence the given salt is CaCO₃ – Calcium carbonate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 10

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

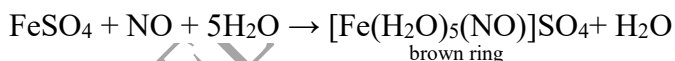
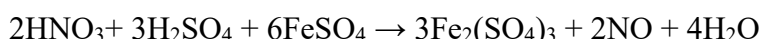
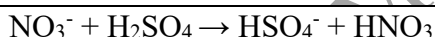
EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	II group acid radical is present. Nitrate (NO ₃ ⁻) may be present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	

CONFIRMATORY TEST FOR NITRATE(NO₃⁻):

Brown ring test: Salt solution in dil.H ₂ SO ₄ +freshly prepared saturated solution of Mohr's salt + conc.H ₂ SO ₄ added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO ₃ ⁻) is confirmed.
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C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	A white precipitate.	V group basic radical present. Ba ²⁺ , Sr ²⁺ , or Ca ²⁺ may be present.
<u>CONFIRMATORY TEST FOR BARIUM (Ba²⁺)</u>		
Original solution + acetic acid + potassium chromate solution.	Yellow precipitate.	Ba ²⁺ is confirmed.
$\text{Ba}^{2+} + 2\text{CH}_3\text{COO}^- \rightarrow (\text{CH}_3\text{COO})_2\text{Ba}$ $(\text{CH}_3\text{COO})_2\text{Ba} + \text{K}_2\text{CrO}_4 \rightarrow \text{BaCrO}_4 \downarrow + 2\text{CH}_3\text{COOK}$		
Flame test: A pinch of salt + few drops conc. HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Apple green colour is imparted to the flame.	Ba ²⁺ is confirmed.

RESULT:

The given salt contains <u>nitrate (NO₃⁻)</u> as acid radical and <u>barium (Ba⁺²)</u> as basic radical.
Hence the given salt is Ba(NO₃)₂ – Barium nitrate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 11

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH ₄ OH.	Group II acid radical is present. Cl ⁻ may be present.
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CONFIRMATORY TEST FOR CHLORIDE (Cl⁻)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.
$\text{Cl}^- + \text{AgNO}_3 \rightarrow \text{NO}_3^- + \text{AgCl} \downarrow$ white ppt. $\text{AgCl} + 2\text{NH}_4\text{OH} \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{H}_2\text{O}$ soluble complex		
2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H ₂ SO ₄ . Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.
$2\text{Cl}^- + \text{K}_2\text{Cr}_2\text{O}_7 + 3\text{H}_2\text{SO}_4 \rightarrow \text{SO}_4^{2-} + \text{K}_2\text{SO}_4 + 2\text{CrO}_2\text{Cl}_2 + 3\text{H}_2\text{O}$ chromyl chloride $\text{CrO}_2\text{Cl}_2 + 4\text{NaOH} \rightarrow \text{Na}_2\text{CrO}_4 + 2\text{H}_2\text{O} + 2\text{NaCl}$ $\text{Na}_2\text{CrO}_4 + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbCrO}_4 \downarrow + 2\text{CH}_3\text{COONa}$ Yellow ppt.		

C| ANALYSIS OF BASIC RADICALS (CATIONS):

<u>TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)</u>		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
<u>DETECTION OF GROUP I BASIC RADICALS</u>		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid + NH ₄ OH solution in excess + (NH ₄) ₂ CO ₃ solution.	A white precipitate.	V group basic radical present. Ba ²⁺ , Sr ²⁺ , or Ca ²⁺ may be present.
<u>CONFIRMATORY TEST FOR BARIUM (Ba²⁺)</u>		
Original solution + acetic acid + potassium chromate solution.	No precipitate.	Ba ²⁺ is absent
<u>CONFIRMATORY TEST FOR STRONTIUM (Sr²⁺)</u>		
Original solution + ammonium sulphate solution, warm.	White precipitate.	Sr ²⁺ is confirmed.
$\text{Sr}^{2+} + (\text{NH}_4)_2\text{SO}_4 \rightarrow \text{SrSO}_4 \downarrow + 2\text{NH}_4^+$		
Flame test: A pinch of salt + few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Crimson red colour is imparted to the flame.	Sr ²⁺ is confirmed.

RESULT:

The given salt contains chloride(Cl⁻) as acid radical and strontium (Sr²⁺) as basic radical.

Hence the given salt is SrCl₂ – Strontium chloride.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 12

A| PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
-	+	

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO ₃ ⁻²) may be present.

CONFIRMATORY TEST FOR CO₃⁻²:

Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Carbonate (CO ₃ ⁻²) is confirmed.
The above gas is passed through lime water.	The lime water turns milky.	
$\text{CO}_3^{-2} + \text{HCl} \rightarrow \text{CO}_2 \uparrow + \text{H}_2\text{O} + \text{Cl}^-$ $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O}$ (milky)		

C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
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DETECTION OF GROUP II BASIC RADICALS

Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
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DETECTION OF GROUP III BASIC RADICALS

Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
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DETECTION OF GROUP IV BASIC RADICALS

Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess + H ₂ S	White precipitate.	IV group basic radical present. Zn ⁺² may be present.
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CONFIRMATORY TEST FOR ZINC (Zn²⁺)

Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts.

i) To one part of the solution add dilute NaOH solution drop by drop till it is excess.

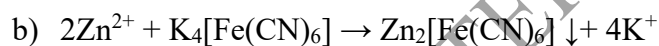
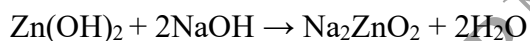
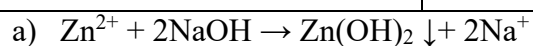
ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.

White precipitate is soluble in excess of NaOH.

White precipitate reappears.

White (or bluish/grayish white) precipitate.

Zn⁺² confirmed.

**RESULT:**

The given salt contains carbonate(CO₃⁻²) as acid radical and zinc (Zn⁺²) as basic radical.

Hence the given salt is ZnCO₃ – Zinc carbonate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 13

A| PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings.Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO₄²⁻):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$\text{SO}_4^{2-} + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{Cl}^-$ white ppt.		

CONFIRMATORY TEST FOR SULPHATE (SO₄²⁻)

Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$\text{SO}_4^{2-} + \text{Pb}(\text{CH}_3\text{COO})_2 \rightarrow \text{PbSO}_4 \downarrow + 2\text{CH}_3\text{COO}^-$ white ppt.		

C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
<u>DETECTION OF GROUP II BASIC RADICALS</u>		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original solution + NH ₄ Cl solid +NH ₄ OH solution in excess. Warm + H ₂ S	Buff precipitate.	IV group basic radical present. Mn ⁺² may be present.
<u>CONFIRMATORY TEST FOR MANGANOUS (Mn²⁺)</u>		
Original solution + NH ₄ Cl solid +NH ₄ OH solution in excess + H ₂ S Add dil.HCl and boil. Add sodium hydroxide solution in excess	Buff / flesh coloured ppt. Precipitate dissolves. White precipitate formed and turns brown on exposure to air.	Mn ²⁺ is confirmed.
$\text{Mn}^{+2} + \text{H}_2\text{S} \rightarrow \text{MnS} + \text{H}^+$ $\text{MnS} + 2\text{HCl} \rightarrow \text{MnCl}_2 + \text{H}_2\text{S}$ $\text{Mn}^{2+} + 2\text{NaOH} \rightarrow \text{Mn}(\text{OH})_2 \downarrow + 2\text{Na}^+$ white ppt. $\text{Mn}(\text{OH})_2 + [\text{O}] \rightarrow \text{MnO}(\text{OH})_2$ Hydrated manganese dioxide (Brown colour)		

RESULT:

The given salt contains sulphate(SO₄⁻²) as acid radical and manganous (Mn⁺²) as basic radical.

Hence the given salt is MnSO₄ – Manganous sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 14

A] PRELIMINARY TEST

7. Nature of the salt: Crystalline
8. Colour of the salt: Colourless
9. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

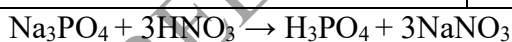
Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO₄²⁻):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	No precipitate	III group acid radical sulphate is absent
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DETECTION OF GROUP III ACID RADICAL(PO₄³⁻):

Salt solution in water + conc. HNO ₃ + solution of ammonium molybdate. Heat to boil.	A canary yellow precipitate is formed.	III group acid radical phosphate is present and confirmed
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C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H ₂ S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH ₄ Cl solid + NH ₄ OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original solution + NH ₄ Cl solid +NH ₄ OH solution in excess. Warm + H ₂ S	White precipitate.	IV group basic radical present. Zn ⁺² may be present.
CONFIRMATORY TEST FOR ZINC (Zn²⁺)		
Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts. i) To one part of the solution add dilute NaOH solution drop by drop till it is excess. ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.	White precipitate is soluble in excess of NaOH. White precipitate reappears. White (or bluish/grayish white) precipitate.	Zn ⁺² confirmed.
c) $\text{Zn}^{2+} + 2\text{NaOH} \rightarrow \text{Zn}(\text{OH})_2 \downarrow + 2\text{Na}^+$ $\text{Zn}(\text{OH})_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + 2\text{H}_2\text{O}$ d) $2\text{Zn}^{2+} + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Zn}_2[\text{Fe}(\text{CN})_6] \downarrow + 4\text{K}^+$		

RESULT:

The given salt contains <u>phosphate(PO_4^{-3})</u> as acid radical and <u>zinc (Zn^{+2})</u> as basic radical.
Hence the given salt is $\text{Zn}(\text{PO}_4)_3$ – zinc phosphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 15

A| PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO ₃
+		

B| ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

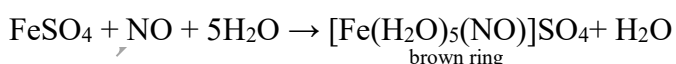
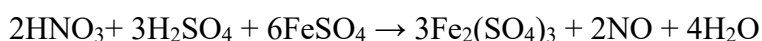
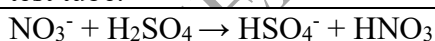
EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H ₂ SO ₄ in a dry test tube.	No reaction	II group acid radical is present. Nitrate (NO ₃ ⁻) may be present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	

CONFIRMATORY TEST FOR NITRATE(NO₃⁻):

Brown ring test: Salt solution in dil.H ₂ SO ₄ + freshly prepared saturated solution of Mohr's salt + conc.H ₂ SO ₄ added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO ₃ ⁻) is confirmed.
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C| ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH₄⁺) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
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PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	White precipitate	I group basic radical present. Pb^{+2} may be present.
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CONFIRMATORY TEST FOR Pb^{+2} RADICAL:

Dissolve the precipitate in hot water and divide the solution into two parts.

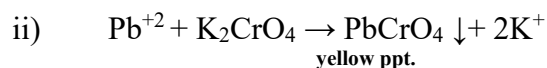
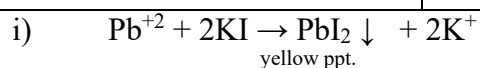
i) Part 1 + KI solution

Yellow precipitate is formed.

Part 2 + potassium chromate solution.

Yellow precipitate is formed.

Pb^{+2} is confirmed.



RESULT:

The given salt contains nitrate(NO_3^-) as acid radical and lead(Pb^{+2}) as basic radical.

Hence the given salt is $Pb(NO_3)_2$ – Lead nitrate.